CW3 SE Report Group 22

Anonymous submission for the SE component of Coursework 3.

1 Task 1: Design Choices and Implementation Notes

1.1 Implemented Use Cases

This submission implements the group-22 main-system use cases:

- Log in
- Log out
- Register entertainment provider
- Create event
- Search for performances
- View performance
- Book performance
- Edit preferences
- Cancel performance
- Cancel booking

1.2 System Structure

Briefly describe the main architecture of the system. For example:

- the main packages and their responsibilities
- the role of controllers, models, view, and external/integration components
- how data is currently stored during execution

Write here:

1.3 Alignment with the Provided UML

State whether the implementation matches the provided class and sequence diagrams exactly. If there were any points where the diagrams could not be followed directly, explain:

- what the mismatch was
- why it could not be implemented exactly as drawn
- what design change was made in code
- how the UML could be updated to stay consistent with the implementation

Write here:

1.4 Key Design Choices

Summarise the most important choices you made when implementing the main system. For example:

- handling of input validation
- object ownership and shared state
- controller responsibilities
- interaction with mock external systems
- error handling strategy in the text interface

Write here:

1.5 Quality Considerations in the Implementation

Briefly mention implementation choices that support quality attributes such as security, privacy, or reliability. This can overlap with the later reflection, but here the focus should be on concrete system decisions.

Write here:

2 Task 5: Reflection

2.1 Reflection on Teamwork

The Experience Our team first established a workflow using Git and GitHub with a fork-and-pull-request model. We also built a CI check action for coursework. For Task 1, we began by dividing the use cases across the team before any implementation. Once the use cases were divided, we wrote placeholder functions in all controller and model files. This approach enabled a unified interface and facilitated assigning specific functions to each team member based on their use cases.

Reflection on Action By using placeholder functions and assigning ownership from the start, we rarely encountered merge conflicts, as we seldom modified the same files or functions simultaneously. This time, our engagement exceeded that of previous coursework, leading to improved output. We also implemented code review via pull requests, which was particularly important. In a complex MVC system with shared data structures and interdependent controllers, this ensured teammates understood each other's APIs and data structures before integrating changes. Our GitHub action CI check verified that CW3 files were modified, that all tests passed, and that the submission ZIP was built, which offers a convenient double-check after work.

Theory We learned that defining function signatures early helps parallel development by reducing conflicts and forcing agreement on interfaces. We also learned that code review is essential in complex systems, as it helps maintain a shared understanding across the team.

Preparation Next time, we will adopt a Kanban-style task board to make the flow of work more visible and surface dependencies between teammates earlier, rather than relying solely on ad-hoc messaging.

2.2 Reflection on the Quality of the Work and Quality Attributes

Suggested structure:

- how well the different tasks were tackled
- one quality attribute shown by the final system
- concrete implementation or testing features that support it
- what could have been improved with more time or resources

Word limit: 250 words maximum

2.3 Reflection on Tools, Agile Approaches and Practices

Suggested structure:

- why each tool, practice, or agile approach was chosen
- what advantages and disadvantages were observed
- what worked well in practice and what did not
- what would be changed next time

Word limit: 250 words maximum

2.4 Screenshots of Tools and Practices

The coursework document explicitly asks for screenshots in the tools/agile/practices reflection. Insert screenshots of the tools you actually used, such as issue tracking, Kanban boards, version control history, CI runs, or planning boards.

Figure 1: Screenshot placeholder 1. Replace with a real screenshot of a tool or practice used by the team.

Figure 2: Screenshot placeholder 2. Replace with another real screenshot if needed.

3 Submission Checklist

Before exporting the PDF, check the following:

- The file is named `CW3SEReportGroup22.pdf`.

- The document is anonymous.
- The declaration of work is not included here.
- Task 1 choices are described.
- All three reflection parts are included.
- Each reflection part is at most 250 words.
- The tools/agile section includes screenshots.